AI based Industrial Anomaly Detection and Root Cause Analysis

By Risa Chaudhari

Motivation

- Industrial machines generate multivariate sensor streams (temperature, pressure, vibration, etc.)
- Small anomalies can cause costly downtime or safety risks
- Traditional monitoring detects faults but cannot explain why or suggest actions

Research Questions

1. **Can we reliably detect anomalies in complex industrial time-series data?**
   (Malhotra et al., 2016; Hundman et al., 2018)

2. **Can causal modeling improve root cause identification and explain anomaly propagation?**
   (Pearl, 2009; Spirtes et al., 2000)

3. **Can combining AI reasoning (LLMs) and temporal validation improve confidence in RCA explanations?**
   (Wei et al., 2022; Yao et al., 2023)

Project Goal

Build an end-to-end explainable pipeline that:

- Detects anomalies
- Identifies root cause
- Tracks propagation path
- Generates human-readable explanations

Output:

- Root cause
- Propagation path
- Confidence score
- Natural language explanation

Dataset

Secure water treatment (SWaT)

- Domain: Industrial water treatment plant
- 51 sensors + actuators across 6 stages
- Normal operation data + simulated attack/fault scenarios
- Contains multivariate time-series for pressure, flow, level, valve positions, etc
- 6 stages - **P1**: Raw water tank, **P2**: Chemical dosing, **P3**: Ultra filtration, **P4**: Dechlorination, **P5**: Reverse osmosis, **P6**: Final storage

System Architecture

Multivariate sensor streams from industrial plant (pressure, flow, level, etc.)

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time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

RCA explanation

System Architecture

time series data

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LLM + temporal validation

RCA explanation

Normalization, windowing, and feature preparation for temporal modeling

System Architecture

time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

RCA explanation

Learns normal behavior and detects anomalies using reconstruction error

System Architecture

time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

RCA explanation

Models causal relationships between sensors for root cause analysis

System Architecture

time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

RCA explanation

Identifies how anomalies propagate across sensors/stages

System Architecture

time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

RCA explanation

Validates causal paths using llm reasoning + time-order consistency

System Architecture

time series data

preprocessing

LSTM - AE

Bayesian Network

Path generation

LLM + temporal validation

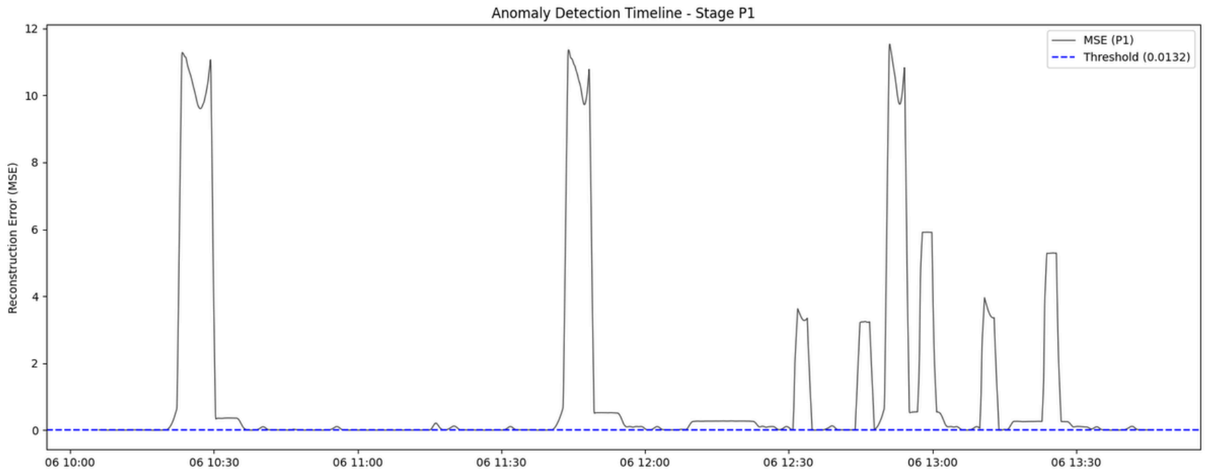
RCA explanation

Generates human-readable root cause and system behavior explanation

Anomaly detection

- Model: **LSTM Autoencoder**
- **6 LSTM AEs for each stage of the plant**
- Trained on normal data only
- Detect anomalies via:

  Reconstruction Error > **Threshold**



- Threshold:
  - 99.9th percentile of normal data
- Window - based anomaly detection: Sliding windows, normalized features (preprocing & windowing)
- Stage-wise anomaly scores - window level

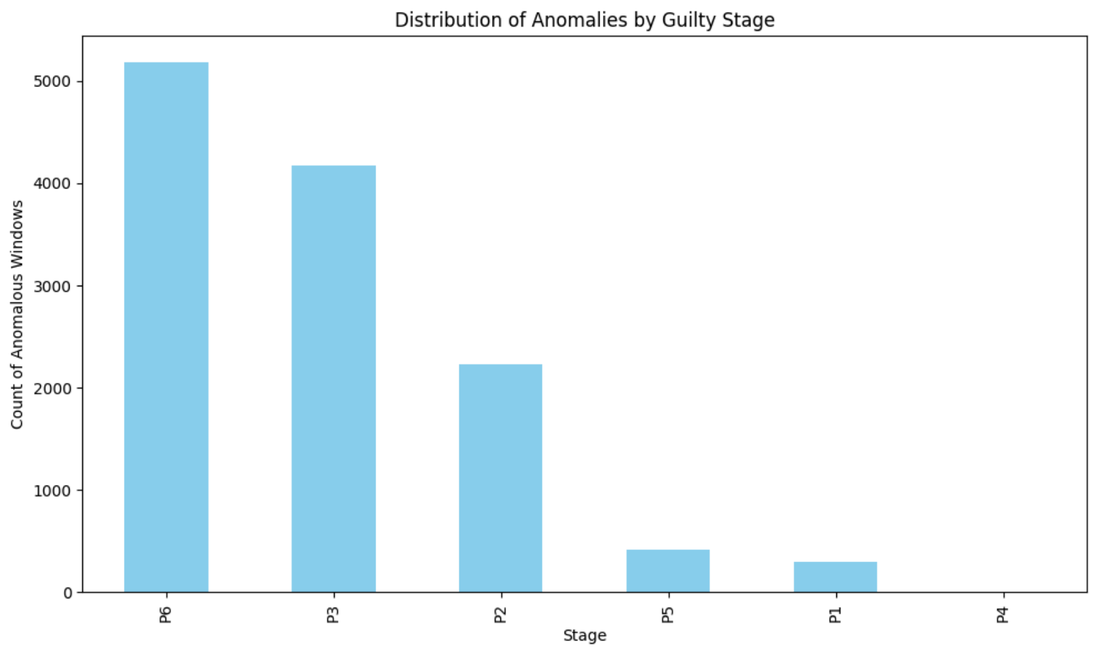
Evaluation Metrics - reconstruction error

Root Cause Analysis

- Stage-level RCA
  - Identify faulty stage
- Feature-level RCA
  - Identify faulty sensor

Learning causal relationships

- Learned stage-wise Bayesian Networks
- Method:
  - Hill Climbing + BIC score
- Nodes: Sensors
- Edges: Causal relationships



Propagation paths

Why BN?

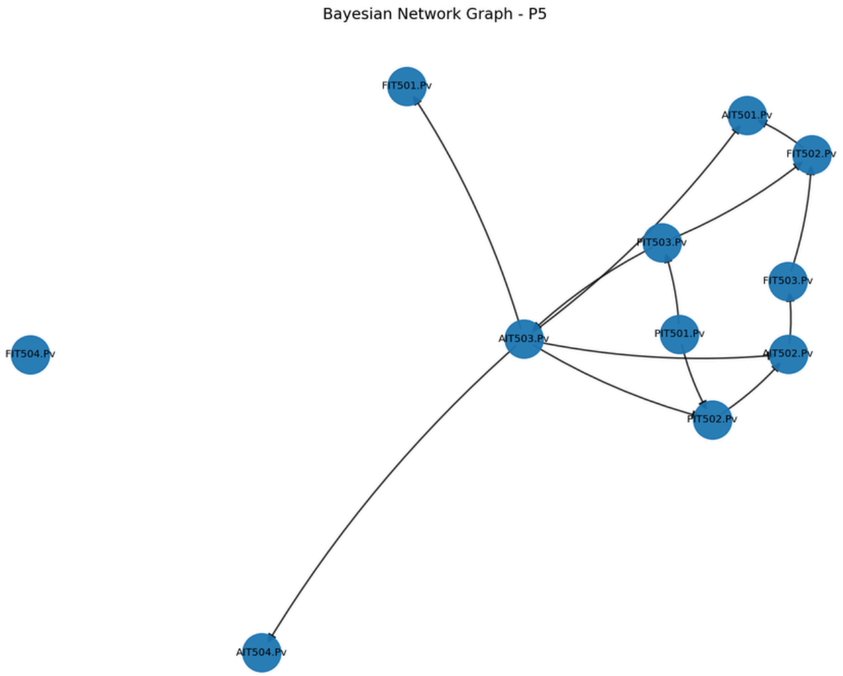
- Captures dependencies
- Enables propagation tracking

Use BN to generate:

- Possible propagation paths

Evaluation Metrics-

- **consistency check (validation)**
- **LLM validation**
- **confidence score**



LLM Evaluation and Temporal Consistency

LLM-based Validation - qwen2.5-7b

LLM checks:

- Sensor coverage
- Causal direction
- Completeness

Output:

- Valid / Invalid
- Confidence score
- Reason

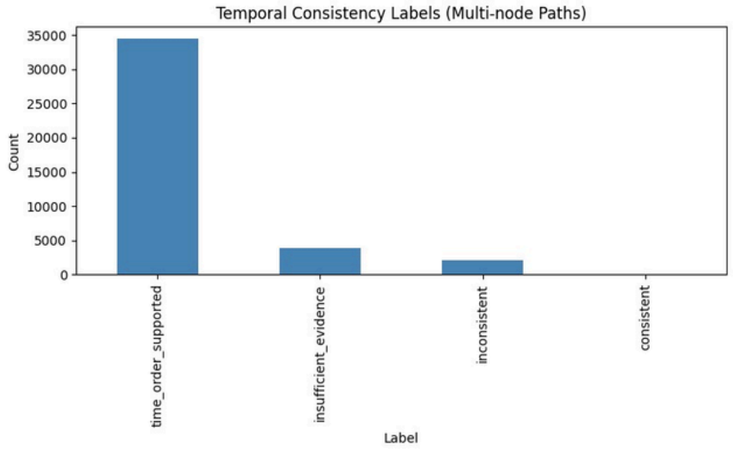
Temporal Validation

Checks:

- Does propagation follow time order?
- Do sensors activate sequentially?

Metrics:

- Temporal score
- Alignment with anomaly time



LLM explanation

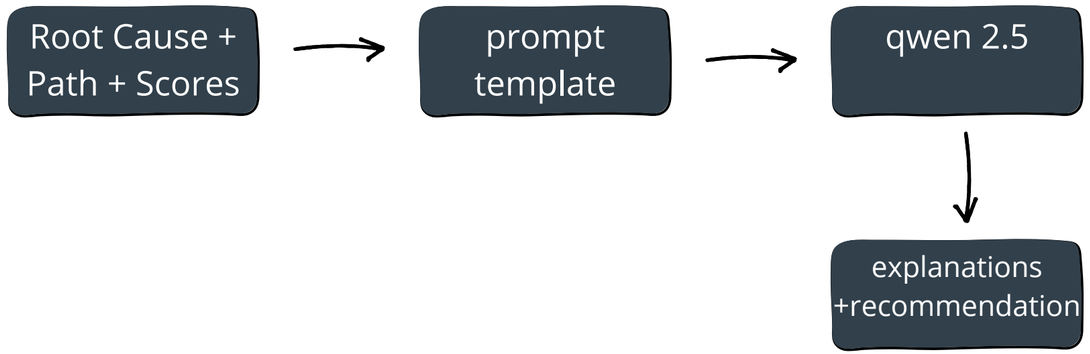
Explainable AI Layer - qwen 2.5-7b

Input:

- Root cause
- Best path
- LLM score
- Temporal score

Output:

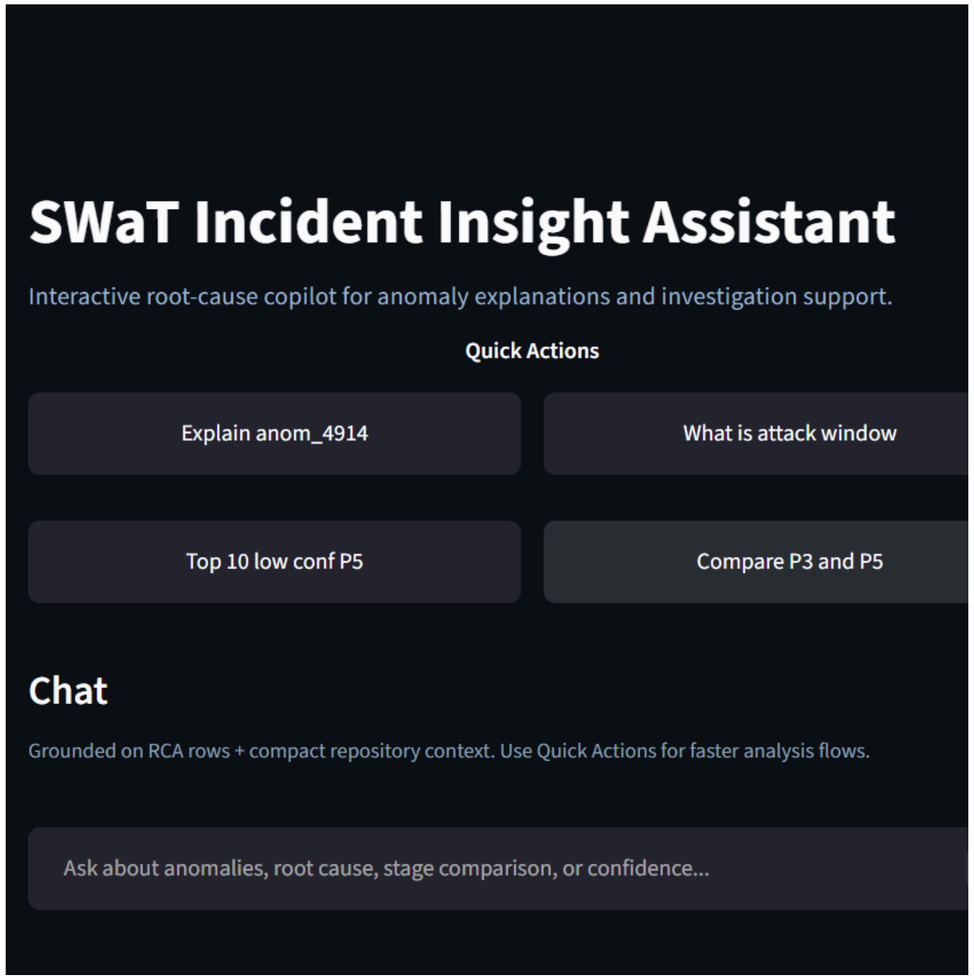
- Summary
- Root cause explanation
- Propagation explanation
- Confidence reasoning
- Recommendation

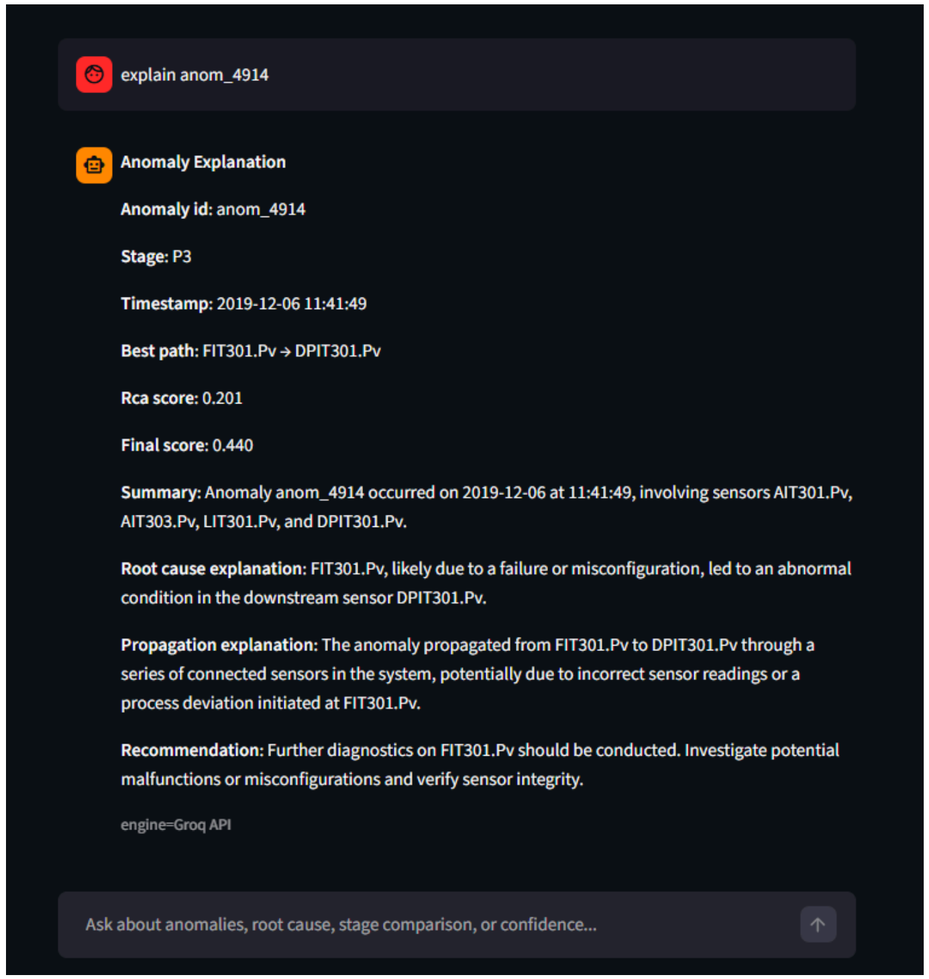


Example Result

- **Anomaly** - ["FIT301.Pv", "DPIT301.Pv", "AIT302.Pv"]
- **Root cause** - [["AIT302.Pv", 0.2009512485136742], ["FIT301.Pv", 0.2009512485136742], ["AIT301.Pv", 0.2009512485136742]]
- **Propagation Path** - FIT301.Pv -> DPIT301.Pv
- **Explanation** - The top root-cause sensors are AIT302.Pv, FIT301.Pv, and AIT301.Pv with scores of 0.2009512485136742 each. However, the evidence is limited, and uncertainty is moderate.The selected propagation path is FIT301.Pv -> DPIT301.Pv. This path covers the anomalous sensors and follows a plausible causal direction from flow to pressure drop, which is coherent within the Ultra-Filtration stage.

Chatbot


SWaT Incident Insight Assistant
Interactive root-cause copilot for anomaly explanations and investigation support.
Quick Actions
Explain anom_4914
What is attack window
Top 10 low conf P5
Compare P3 and P5
Chat
Grounded on RCA rows + compact repository context. Use Quick Actions for faster analysis flows.
Ask about anomalies, root cause, stage comparison, or confidence...


explain anom_4914
Anomaly Explanation
Anomaly id: anom_4914
Stage: P3
Timestamp: 2019-12-06 11:41:49
Best path: FIT301.Pv → DPIT301.Pv
Rca score: 0.201
Final score: 0.440
Summary: Anomaly anom_4914 occurred on 2019-12-06 at 11:41:49, involving sensors AIT301.Pv, AIT303.Pv, LIT301.Pv, and DPIT301.Pv.
Root cause explanation: FIT301.Pv, likely due to a failure or misconfiguration, led to an abnormal condition in the downstream sensor DPIT301.Pv.
Propagation explanation: The anomaly propagated from FIT301.Pv to DPIT301.Pv through a series of connected sensors in the system, potentially due to incorrect sensor readings or a process deviation initiated at FIT301.Pv.
Recommendation: Further diagnostics on FIT301.Pv should be conducted. Investigate potential malfunctions or misconfigurations and verify sensor integrity.
engine=Groq API
Ask about anomalies, root cause, stage comparison, or confidence...

Limitations

Data Dependency

- Relies on normal operating data quality
- Performance may degrade with noisy or incomplete data

Generalization

- Models are trained on a specific plant (SWaT)
- May not directly generalize to other industrial systems

Propagation Path Uncertainty

- Multiple possible paths may exist
- Difficult to guarantee true root cause vs correlated signals

Conclusion

1. **(RQ1) Can we reliably detect anomalies in complex industrial time-series data?**
- LSTM-AE (unsupervised): Learn normal patterns, detect via reconstruction error
- Multi-stage detection: Per-feature loss → anomaly windows
2. **Can causal modeling improve root cause identification and explain anomaly propagation?**
- Bayesian Networks: Model dependencies, rank root causes
- Propagation paths: Trace anomaly spread
3. **Can combining AI reasoning (LLMs) and temporal validation improve confidence in RCA explanations?**
- LLM reasoning: Validate causal paths
- Temporal checks: Ensure correct sequence, improve confidence

- **The results are integrated with a chatbot**

AI usage disclosure

Level 3 — Substantive Research Assistance

Use cases:

- AI-generated code used in experiments
- Data analysis assistance
- Model design suggestions

Key rule: Transparency + reproducibility required.

Declaration (Methods section + Acknowledgments): Generative AI tools were used in the development of [code/analysis/design]. The authors validated all outputs and ensured reproducibility.

References

1. Malhotra et al. (2016) LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection
2. Hundman et al. (2018) Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Thresholding
3. Pearl (2009) Causality: Models, Reasoning, and Inference
4. Spirtes et al. (2000) Causation, Prediction, and Search
5. Wei et al. (2022) Chain-of-Thought Prompting Elicits Reasoning in Large Language Models
6. Yao et al. (2023) ReAct: Synergizing Reasoning and Acting in Language Models
7. Dataset SWaT

Thank you